



Introduction to JavaScript: What, Why, Where, and How to Run

Let's get started



CONTENT

01 Physical Activity

02 Basics of JavaScript

03 JavaScript Application Areas

04 How to Run JavaScript





01

Physical Activity

Activity Name: "You Are the JavaScript"



Person with no "Brain"

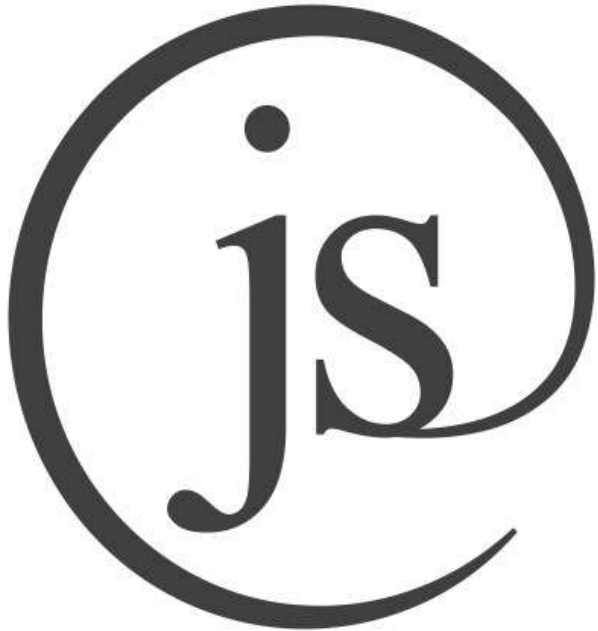
Let's play game to understand what JavaScript is..

02

Basics of JavaScript



What is JavaScript?



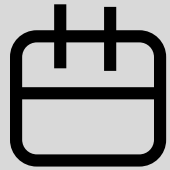
- **Definition and History**

JavaScript is a high- level, dynamic, and interpreted programming language primarily used for enhancing interactivity in web pages. Initially created in 1995, it has evolved significantly, becoming a core technology of the World Wide Web.

- **Core Features**

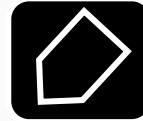
Key features of JavaScript include its event- driven nature, prototype- based object orientation, and the ability to manipulate the Document Object Model (DOM), enabling developers to create responsive and interactive user interfaces.

Why Learn JavaScript?



Popularity and Demand

JavaScript is one of the most widely used programming languages globally, with a vast job market. Its demand continues to grow as more companies adopt it for front- end and back- end development.



Versatility in Web Development

JavaScript is essential for both client-side and server- side programming, making it a versatile choice for developers. Its frameworks and libraries, like React and Node.js, further enhance its utility in web applications.



Community and Resources

A vibrant community surrounds JavaScript, providing extensive support, tutorials, and resources. Platforms like GitHub and Stack Overflow are valuable for learning and troubleshooting, ensuring developers can find help easily.



03

Basics of JavaScript



Front-End Development

Enhancing User Interfaces



JavaScript enables developers to improve user interfaces with dynamic elements, animations, and responsive designs that provide a richer user experience on websites.

Creating Interactive Web Pages



With JavaScript, developers can create interactive features such as sliders, pop-ups, and form validations, making web pages more engaging and user-friendly.

Back-End Development



Server-Side Programming

JavaScript can be utilized for server-side programming, allowing developers to handle application logic and database interactions securely and efficiently.



Using Node.js

Node.js, a runtime environment for JavaScript, enables developers to build scalable server applications that handle multiple connections simultaneously, enhancing performance.

Full Stack Development



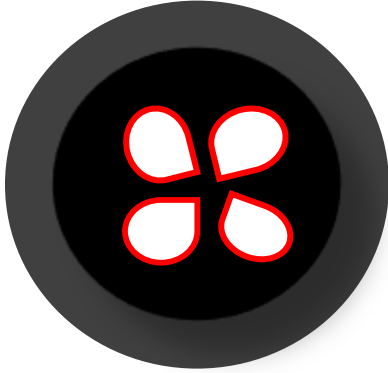
Full stack development combines both front- end and back- end using JavaScript frameworks and libraries, empowering developers to create complete web applications with seamless integrations.

04

How to Run JavaScript



Running JavaScript in Browsers



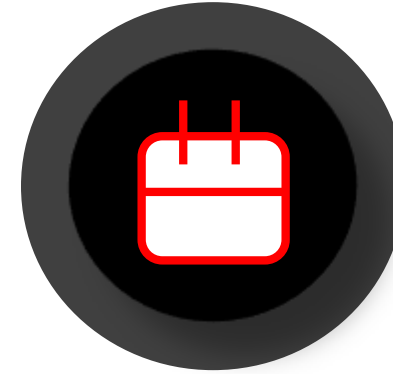
Browser Consoles

Browser consoles allow developers to execute JavaScript code snippets directly within the browser environment, providing instant feedback and debugging capabilities.



HTML Script Tags

HTML script tags are used to embed JavaScript code within HTML documents, enabling dynamic content manipulation when the page loads or events occur.



Development Tools

Modern browsers come equipped with development tools that facilitate JavaScript debugging, performance monitoring, and inspection of elements, enhancing the developer experience.

Environment setup in VS Code

